Lesson plan · for the teacher

Break it down

Programming · Level 1 · Lesson 4

Time
40 min

Materials
Board and chalk; Paper for maps and algorithms; Your phone (optional) with this lesson downloaded

Learning goals

- Name the four parts of computational thinking.
- Compare two algorithms by counting steps.
- Plan a computational-thinking moment in your subject.

Activity: Guess my number

1. Warm-up (5 min): Think of a number from 1 to 100. Pupils guess; you say “higher” or “lower”. Tally the guesses on the board.
2. Teach (10 min): Name the four parts: decomposition, patterns, abstraction, algorithms. Give one school example of each and ask pupils for another.
3. Halving (10 min): Teach “guess the middle, cross out half”. Play again in pairs and compare the counts with the warm-up. Ask why halving needs the numbers in order.
4. Plan (10 min): Groups take a task from another subject (a recipe, a journey, an experiment) and find the four parts in it.
5. Wrap-up (5 min): Ask which part was hardest to spot, and where else halving helps. Then give the exit ticket.

Keep the board tally of guesses: the drop from the warm-up to halving is the lesson’s best evidence.

Discussion

- Why does halving need the numbers in order?
- Which part of computational thinking do you already teach without naming it?
- Where else could we compare two ways of doing a job by counting steps?

Exit ticket

- Name the four parts, with one example of each from today.

Try it in class tomorrow
Play “guess my number” from 1 to 100. Let pupils guess freely and count the guesses, then teach halving. Can anyone beat 7 guesses?

Worksheet · for pupils

Break it down

Programming · Level 1 · Lesson 4

Name _____ Class _____ Date _____

1 **Which of these is decomposition?**

- ◯ A seating plan that shows only the names
- ◯ Noticing that every song has a chorus
- ◯ Splitting a class play into scenes, costumes and music

2 **Match each part of computational thinking to a school example.**

Decomposition	• • • •	A bus map that shows only the stops
Pattern recognition	• • • •	The steps of long division
Abstraction	• • • •	Splitting a science project into tasks for each group
Algorithm	• • • •	Noticing that every multiple of 5 ends in 0 or 5

3 **This short program makes the first guess in "guess my number" from 1 to 100. What does it print? (// divides and drops the remainder.)**

```
low = 1
high = 100
guess = (low + high) // 2
print("Is it", guess)
```

- ◯ Is it 100
- ◯ Is it 1
- ◯ Is it 50

4 **Put the steps of the halving strategy in order.**

- ☐ Hear "higher" or "lower"
- ☐ Guess the middle number
- ☐ Repeat with the half that is left
- ☐ Cross out the half it can't be in

5 **With halving, about how many guesses can it take to find a number from 1 to 1,000?**

- ◯ About 10
- ◯ About 100
- ◯ About 500

Answers · for the teacher

Break it down

Programming · Level 1 · Lesson 4

1 **Splitting a class play into scenes, costumes and music**

Breaking a big job into smaller parts is decomposition.

2 **Decomposition — Splitting a science project into tasks for each group**
Pattern recognition — Noticing that every multiple of 5 ends in 0 or 5
Abstraction — A bus map that shows only the stops
Algorithm — The steps of long division

Breaking into parts, spotting what repeats, keeping only what matters, and exact steps: the four parts appear in every subject you teach.

3 **Is it 50**

(1 + 100) // 2 is 101 // 2, which is 50: the middle.

4 **Top to bottom: 2, 1, 4, 3**

Each guess at the middle throws away half of what is left, which is why so few guesses are needed.

5 **About 10**

Each guess halves what is left: 1,000, 500, 250 … down to 1 in about 10 steps.

For you to think about

In which topic you teach next week could pupils use decomposition, patterns or abstraction? Write the moment, in two or three sentences.